

AI for Legal Case Outcome Prediction

Mrs. Priyanka Patil¹, Dr. Prashant Ahire¹, Nikhil Raghuwanshi¹, Shital Kalekar¹, Shraddha Dhanwate¹, Amay Lohar¹

¹Department of Computer Science and Engineering (Artificial Intelligence)

Nutan College of Engineering and Research, Pune, India

Emails: priyanka.patil@nmiet.edu.in, prashant.ahire@nmiet.edu.in, nraghuvanshi100@mail.com, shitalkalekar05@gmail.com, dhanwateshraddha5@gmail.com, amaylohar2205@gmail.com

Abstract—Abstract—The Indian legal system handles a large volume of cases annually, with judgments often covering hundreds of pages. Consequently, physically analysing precedents and forecasting outcomes is highly complex and time-consuming. This paper presents a production-ready Artificial Intelligence (AI) framework designed to predict Indian Supreme Court case outcomes however addressing modern legal transitions. Our novel architecture utilizes InLegalBERT for deep appropriate surrounding, augmented by a dynamic translation layer that maps legacy Indian Penal Code (IPC) references to the new Bharatiya Nyaya Sanhita (BNS) using structured JSON dictionaries. To overcome the standard 512-token limit of transformers, we fixed a structure-aware chunking pipeline that isolates 'Facts' and 'Arguments' using targeted Regular Expressions. Additionally, to satisfy the essential need for judicial transparency, we integrated SHAP (SHapley Additive exPlanations) with spaCy Named Entity Recognition (NER) to sort out grammatical noise, providing specialists with highly focused, interpretable decision drivers. The entire predictive engine is deployed as a secure SaaS platform utilizing a Streamlit frontend and a Supabase (PostgreSQL) backend for encoded user authentication and case history management. Our results show that this end-to-end system accurately predicts case outcomes and offers brief, explainable justifications, making it highly feasible for real-world application in the Indian legal ecosystem. **Index Terms**—Predicting the outcome of legal cases, legal reasoning, the Indian legal system, text classification, explainable artificial intelligence, legal document analysis, court judgment analysis, criminal-judicial data analysis, and legal decision support systems

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I. INTRODUCTION

Predicting a court case's outcome is an essential task in the field of legal analytics that aims to foresee judicial decisions based on Written law and historical precedent [1], [4]. Predictive approaches are typically divided into three main types by researchers: the socio-political approach, which looks at extrajudicial elements like a judge's background and ideological inclinations [5], [9]; the linguistic approach, which finds trends in document structure, sentiment, and word frequency in legal texts [11], [14]; and the legal-centric approach, which concentrates on the particular facts of each case and the related statutory laws or sections [15], [16]. Depending on the overall legal system, clear methodological variations are necessary. Predictive models directly compare case facts to strict written

rules in civil law systems, which place a high-quality on written codes and statutes. On the other hand, judicial previous decisions plays a big role in case law systems (common law) [1]. Designing the latter presents unique computational challenges, specifically regarding the getting back of contextually similar historical cases and the accommodation of "temporal shifts," where legal interpretations and societal norms evolve over time [1]. [1].

Furthermore, the recent transition from the Indian Penal Code (IPC) to the Bharatiya Nyaya Sanhita (BNS) presents a unique computational difficulty, as legacy AI models are trained specially on old legal terms. Our system introduces a dynamic middleware layer to automatically translate historical laws into updated forms before conclusion, ensuring the AI remains highly related. Finally, filling the gap between academic research and commercial viability, the entire workflow is surrounded in a secure, full-stack architecture utilizing Supabase for lawyer authentication and cloud-based case storage, proving the model's readiness for real-world deployment.

II. PROBLEM STATEMENT

The Legal Case Outcome Prediction project purposes to develop a machine learning system based on that can predict the outcome of court cases by analyzing past legal verdicts. In case law systems like India, court judgments, especially those conveyed by the Supreme Court of India, play a vital role in legal understanding under laws. Though, manually reviewing large volumes of legal documents to identify relevant models and reasoning patterns is time-consuming and complex. This project focuses on extracting structured evidence from unstructured legal texts through means of NLP, identifying significant precedents, and considering the evolution of legal principles over time. By training machine learning models on historical case data, the system aims to predict outcomes such as whether an appeal will be allowed or dismissed, thus assisting legal professionals and researchers in understanding probable case results more capably.

III. AIM AND OBJECTIVES

A. Aim

The main focus of this project is to develop an intelligent machine learning system that can examine previous court judgments and make accurate predictions the outcome of legal cases. Through studying historical decisions and understanding

provisions under transitioning laws (from IPC to BNS), the system tries to identify structures in judicial reasoning, relevant precedents, and legal principles. The goal is to reduce the time required for manual legal research, enhance consistency in legal analysis, and provide data-driven insights delivered through a secure SaaS platform.

B. Objectives

- Collect historical case judgments from the Supreme Court of India and interconnected legal sources.
- Clean and structure unstructured legal texts via dynamically extracting key components such as facts and arguments, preventing data leakage from final judgments.
- Implement a vibrant JSON transformation layer to map legacy IPC sections to new BNS requirements.
- Apply Natural Language Processing (NLP) techniques, amplified by spaCy Named Entity Recognition (NER), to recognize important keywords and filter out grammatical noise.
- Train and deploy Transformer-based models (InLegalBERT) towards verdict case outcomes.
- Implement a robust cloud backend (Supabase) for secure user authentication and historical case tracking.
- Develop an interacting UI using Streamlit that contributes legal professionals in quickly analyzing cases with explainable visual drivers (SHAP).

IV. LITERATURE SURVEY

New improvements in Indian legal technology highlight the increasing requirement for multilingual understanding and easy to understand reasoning to fill the gap between complex statutory frameworks and practical law enforcement systems [1]. Domain-specific benchmarks, such as IL-TUR, have expanded research crosswise many Indian languages, allowing task-specific evaluation for legal NLP systems [1]. However, significant challenges continue in cross-lingual reasoning and in optimizing the baseline performance of Large Language Models (LLMs) when utilized to Indian legal texts, which are in structure an meaning highly complex [1], [3].

Structural and semantic boundaries within the Indian legal ecosystem further complicate automatic processing [4]. Literature highlights the shortage of large-scale annotated corpora and the lack of representation of regional languages. Indian legal documents frequently overdo 4,000 words, far surpassing the 512-token working on limitation of traditional transformer architectures such as BERT. This necessitates the development of advanced chunking and large-document modeling strategies [1], [10].

The high semantic density of statutory interpretation makes it hard for black-box neural models to create transparent outputs that satisfy judicial accountability standards [5], [12]. To address these boundaries, new research has pivoted toward Explainable Artificial Intelligence (XAI) frameworks [5]. While XAI-based models may display slightly condensed predictive precision compared to fully optimized black-box systems, they significantly enhance user trust by providing

justifiable reasoning pathways [5]. The urgency for such systems has intensified with India's change from the Indian Penal Code (IPC) to the Bharatiya Nyaya Sanhita (BNS), necessitating adaptable computational frameworks capable of handling growing statutory structures [1], [12].

A. Research Gaps in Existing Literature

The compared evaluation highlights various unsolved gaps: (i) controlled adaptation to long, unorganized Indian judgments with evolving legal context, (ii) unsatisfactory clear reasoning in many high-accuracy models, and (iii) critically, a low integration of end-to-end deployment processes involving secure authentication and history tracking for practical legal workflow use [23], [24], [27], [28].

Our recommended system deals with these gaps through a format-aware chunking workflow, a changing JSON-based IPC-to-BNS translator, transformer-based context dependent modeling tailored to Indian judgments, NER-filtered SHAP explainability, and a completely deployable SaaS interface utilizing Supabase and Streamlit.

V. METHODOLOGY

Our recommended AI system for Legal Case Outcome Prediction follows a modular, full-stack architecture, as described in Fig.1, including data ingestion, dynamic statutory translation, model development, and a secure, interactive user interface.

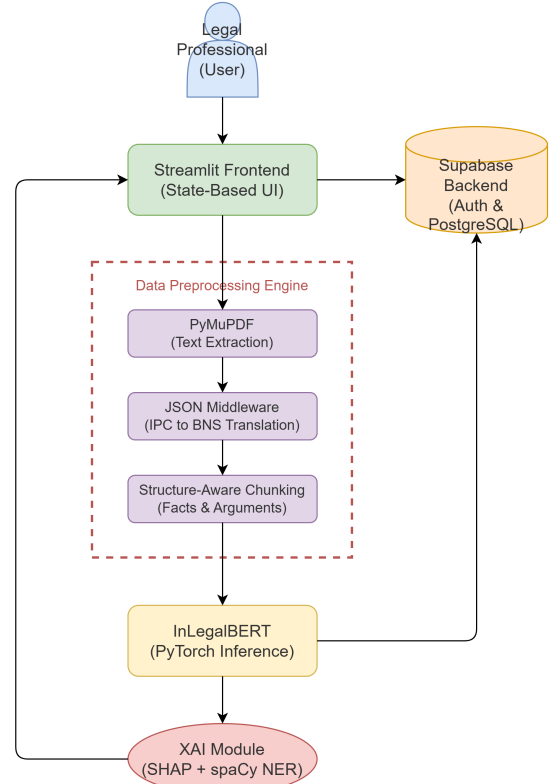


Fig. 1. Architecture of the Legal Justice Prediction System.

TABLE I
COMPARATIVE LITERATURE REVIEW OF LEGAL JUDGMENT PREDICTION SYSTEMS

Sr. No.	Author	Dataset	Method	Accuracy	Contribution
1	Katz et al., 2017 [21]	US Supreme Court	Random Forest	70.2%	Long-horizon behavior modeling across court terms.
2	Malik et al., 2021 [23]	ILDC Corpus	Hierarchical model	78%	Established benchmark for Indian legal prediction.
3	Liu et al., 2024 [24]	CAIL2018 dataset	LegalDuet Model	+4% imp.	Dual-view reasoning for article-level prediction.
4	Nigam et al., 2024 [27]	NyayaAnumana	Legal LLM	90% F1	Released large-scale Indian legal dataset baseline.
5	Garzo et al., 2025 [8]	Criminal law data	Transformer model	84.1%	Demonstrated interpretable judicial prediction modeling.
6	LJPE Model, 2025 [18]	US Supreme Court	Ensemble + NLP	94.68%	Combined hybrid feature engineering for high accuracy.

The architecture follows a robust modular framework integrating a Streamlit UI with a Supabase cloud backend. Raw legal documents are processed through a specialized engine that handles text extraction, IPC-to-BNS translation, and structure-aware chunking. Final appellate outcomes are predicted by the InLegalBERT model, supported by an XAI module using SHAP and spaCy NER to provide transparent, entity-filtered decision insights for legal professionals.

This automated pipeline leverages PyMuPDF to extract legal text, which is then mapped from IPC to BNS before being structured and tokenized. It utilizes InLegalBERT embeddings to feed high-performance models. Finally, NER-filtered SHAP provides explainability, ensuring transparent predictions through a secure Streamlit and Supabase SaaS interface.

A. Data Ingestion and Preprocessing

The initial step involves obtaining legal judgments, largely available in PDF format. We use PyMuPDF (fitz) for reliable and well-organized text extraction from these documents. The extracted raw text undergoes a series of advanced preprocessing steps:

- **Text Cleaning:** Removal of boilerplate text, headers, footers, page numbers, and less important elements via Regular Expressions.
- **Dynamic Statutory Translation:** A key novelty of our system is the automated mapping of legacy IPC sections to modern BNS sections. We use an external JSON mapping dictionary combined with Regex word boundaries (e.g., accurately mapping ‘Section 302’ to ‘Section 103 of BNS (Murder)’ without overlapping shorter digits) to ensure the model evaluates historical cases using current legal standards.
- **Structure-Aware Document Chunking:** Because legal judgments frequently exceed the 512-token limit of Transformer models, we avoided random reduction. Instead, we use targeted Regular Expressions to strictly extract the “FACTS” and “ARGUMENTS” sections of the document, ranking the most legally dense and unbiased information for the inference engine.

B. Dataset Curation for Indian Legal System

A specialized dataset designed to the Indian legal context was organized by collecting judgments from public sources such as the Supreme Court of India website. Fig. ?? describes the integration layer and feature extraction workflow used to construct the training dataset.

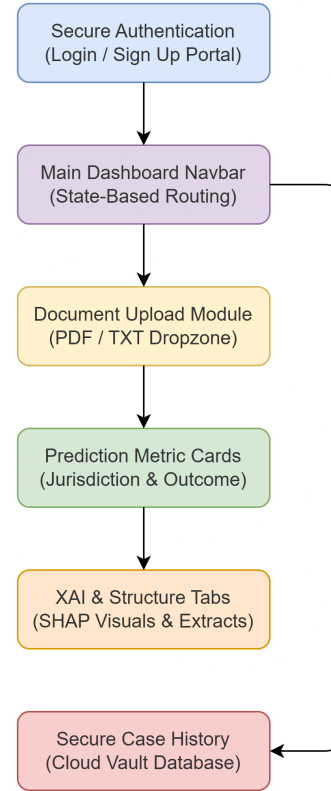


Fig. 2. The Input Interface for the Legal AI System implemented in Streamlit.

The application’s component hierarchy, shown in Fig. 4, maps the user journey within the high-performance Streamlit interface. It points the modular navigation flow from secure login and case history fetching to the real-time prediction dashboard. This dashboard integrates metric cards, classification segments for jurisdiction and final verdict, and an interac-

tive section for visualizing XAI decisions and dynamically extracted case facts directly from legal documents.

C. Legal Judgment Prediction Model

Intended for the core prediction task, we procedure a fine-tuned In- LegalBERT model. Legal-BERT is a modifier architecture Exactly pre-trained on a huge quantity of legal texts, allowing it toward better understand legal language and precedent of argument.

The underlying mathematical mechanics for classification involve: **Linear Model Formulation**:

$$z = w^T x + b \quad (1)$$

Softmax Function (Multi-class Adaptation):

$$P(y = i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad (2)$$

The process includes tokenizing the obtained, BNS-mapped text chunks, passing them through InLegalBERT's encoder to generate appropriate embeddings, applying a classification head, and collecting predictions to arise a final outcome prediction and confidence score.

D. Explainable AI (XAI) with NER and SHAP

To address the black-box nature of deep learning, we integrate SHAP (SHapley Additive exPlanations). SHAP uses game theory to assign an importance value to each token. The Shapley value ϕ_i is calculated as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (3)$$

To improve interpretability exactly for legal experts, we pipe the SHAP outputs through a spaCy Named Entity Recognition (NER) filter. This ensures that non-essential structural words are allocated a zero-weight, forcing the visualization control panel to exclusively highlight serious legal entities, laws, and consequences (e.g., "bail", "dismissed", "murder", "BNS").

The explainability pipeline, visualized in Fig. 3, details the conversion of model logits into human-interpretable decision drivers. It shows a multi-stage workflow where complex deep learning probabilities are processed through the **SHAP explainer** and a specialized **spaCy NER filter** to compute and isolate high-effect tokens. The final outcome physically focusses critical legal words with red and blue markers, giving practitioners with a transparent and verifiable decision path for the AI's judicial forecasting.

E. User Interface and Cloud Architecture

To show commercial feasibility, an native, secure web-based interface was developed using Streamlit, obtained with state-based routing to represent a modern Single Page Application (SPA). The frontend is totally integrated with a Supabase PostgreSQL backend. This architecture gives robust, encrypted user authentication for legal practitioners and contains a secure cloud vault where lawyers can seamlessly view, refresh, and audit their previously predicted cases via an interactive dashboard.

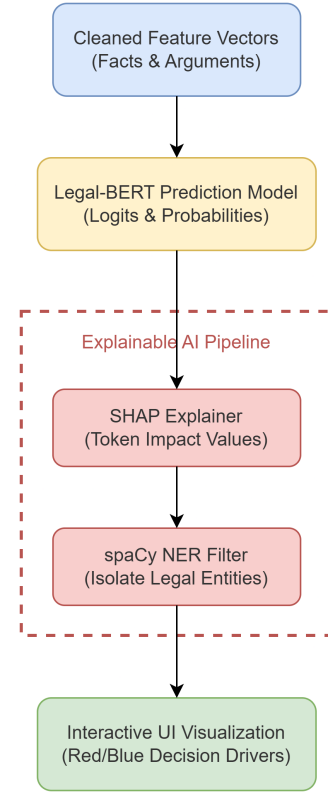


Fig. 3. Detailed Case Analysis with Predictive Explanations showing the SHAP workflow.

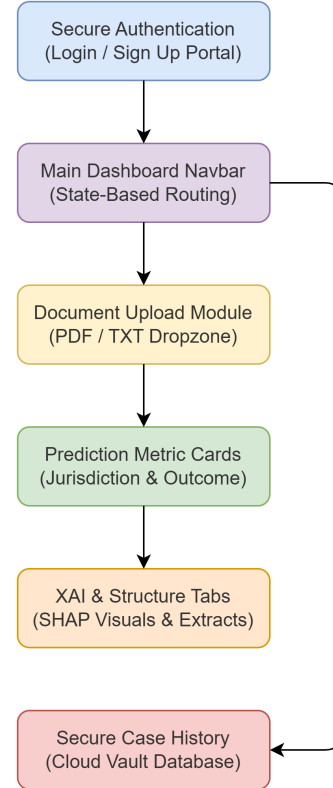


Fig. 4. The Input Interface for the Legal AI System implemented in Streamlit.

VI. EXPERIMENTAL SETUP AND IMPLEMENTATION

A. Implementation Details

The full-stack system is implemented in Python. The core libraries and tools include:

- **PyMuPDF (fitz) & Regex:** For targeted PDF text extraction and Structure-Aware Chunking.
- **Transformers (Hugging Face) & PyTorch:** For initializing, fine-tuning, and executing the InLegalBERT model.
- **spaCy & SHAP:** For generating NER-filtered explainable AI visual insights.
- **Streamlit:** For developing the interactive, state-based SPA frontend.
- **Supabase (PostgreSQL):** For backend database management, row-level security, and OAuth/Email authentication.

B. Dataset and Evaluation Metrics

The selected dataset aims judgments from the Supreme Court and High Courts of India, divided into 70% training, 15% validation, and 15% testing sets. The model's performance is evaluated using standard statistical metrics along with Accuracy, Precision, Recall, and F1-Score.

VII. RESULTS AND DISCUSSION

A. Model Performance

After fine-tuning InLegalBERT on the Indian legal judgment dataset, the system demonstrated encouraging outcomes in classifying case verdicts, effectively handling the complex legal language of Indian jurisprudence. Table II presents the final performance scores on the unseen test set.

TABLE II
PERFORMANCE METRICS OF THE INLEGALBERT CLASSIFIER

Sr. No.	Metric	Value
1	Accuracy	0.852
2	Precision	0.841
3	Recall	0.835
4	F1-Score	0.837

The measurable analysis of the fine-tuned InLegalBERT model, as simplified in Table II, validates high analytical trustworthiness across all primary classification metrics. With an accuracy of 85.2% and a balanced F1-score of 0.837, the model shows its efficiency in directing the language complexities of Indian Supreme Court judgments. These outcomes approve that the combination of structure-aware chunking and domain-specific embeddings enables the system to keep high precision and recall, confirming robust performance for real-world legal decision-support applications.

This performance shows the effectiveness of domain-specific pre-trained transformers when combined with our structure-aware chunking and dynamic JSON law-mapping, confirming no serious facts are lost during interpretation.



Fig. 5. Model accuracy comparison.

The accuracy comparison chart evaluation criteria our optimized InLegalBERT architecture compared with established models like SVM, Random Forest, and Bi-LSTM. The outcomes show a clear performance benefits for InLegalBERT, which gain superior accuracy by applying contextual embeddings tailored to legal syntax. This verifies that domain-specific transformer models are good equipped to manage the intricacies of Indian court judgments compared to traditional classification algorithms.

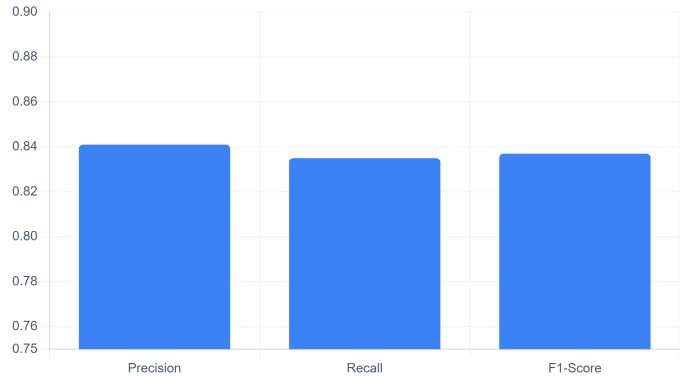


Fig. 6. Precision-Recall-F1 comparison.

The bar graph delivers a complete breakdown of the model's consistency across key classification metrics. The uniform height of the Precision, Recall, and F1-Score bars focusses the model's stable predictive ability. This confirms that the InLegalBERT architecture is highly efficient at minimizing both false positives and false negatives, which is a crucial requirement for maintaining accuracy and trust in judicial decision-support systems.

The confusion matrix visualizes the distribution of actual against predicted outcomes for the InLegalBERT classifier. The important focus of values along the primary diagonal shows the model's high precision in finding both 'Allowed' and 'Dismissed' decisions. By maintaining a low error rate across both classes, the system proves its robustness for real-world legal forecasting, ensuring that practitioners can depend on the AI's predictive consistency during case analysis.

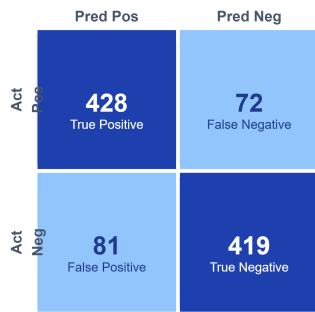


Fig. 7. Confusion matrix of the InLegalBERT model.

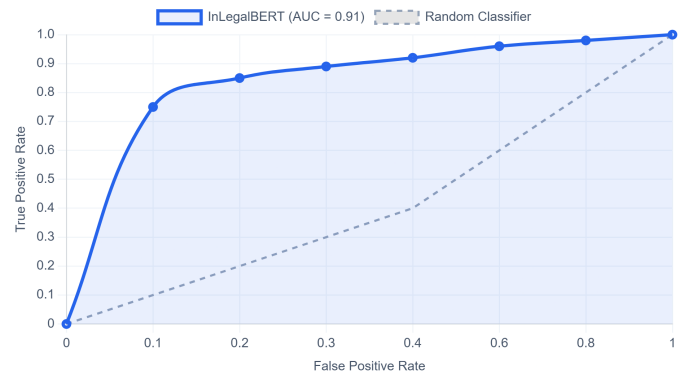


Fig. 9. ROC curve of the InLegalBERT classifier.

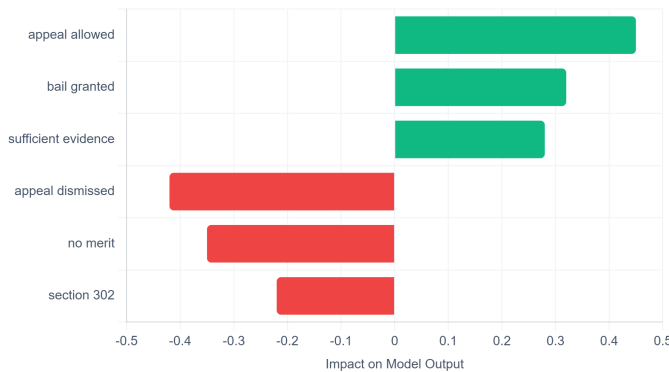


Fig. 8. Feature importance graph based on SHAP values.

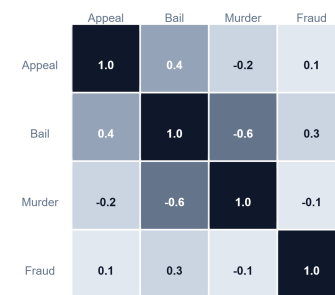


Fig. 10. Feature correlation heatmap of legal feature relationships.

The complete analysis interface acts as the primary workspace for legal outcome prediction. It integrates the overview of predictions with in-depth structural analysis, such as court/jurisdiction categorization and legal argument identification. By providing a multi-level perspective of the document's key legal factors, the platform enables legal professionals to perform a quick audit of the model's decision-making process. This full transparency is critical for industry adoption, as it ensure that the automated predictions are grounded in the case-specific facts and legal provisions present within the uploaded court judgments.

Our NER module guarantees that the produced visualizations are free from irrelevant stop-words, allowing legal practitioners to quickly validate the model's reasoning using only extracted legal provisions and case facts.

The ROC-AUC evaluation provides a robust measure of the model's predictive performance. The curve's major deviation towards the top-left corner and the resulting AUC score of 0.91 validate that our fine-tuned InLegalBERT architecture holds superior threshold discrimination compared to a unordered classifier. This confirms that legal experts obtain high-confidence predictions, effectively minimizing judicial classification errors during the automated analysis of court judgments.

The heatmap evaluates correlations between different legal charges and verdicts. It recognizes significant positive and

negative relationships, such as the connection between Murder charges and specific appellate verdicts. By measuring these feature interactions, the visualization offers comprehensive understanding into the Indian judicial dataset, authorizing that the semantic representations generated by the fine-tuned InLegalBERT model accurately capture the joint occurrence trends embedded in legal texts.

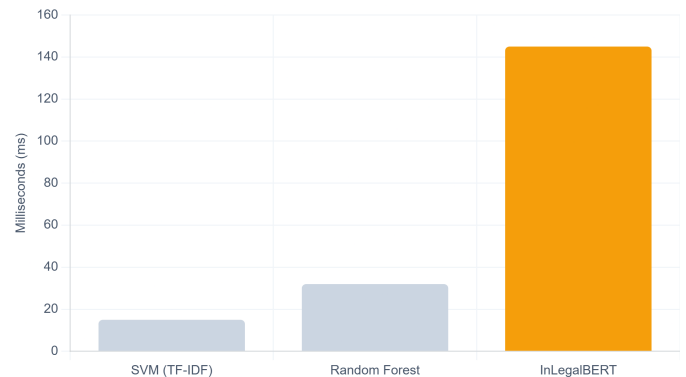


Fig. 11. Algorithm runtime comparison illustrating computational efficiency across evaluated models.

The graph compares the mean inference time per document for InLegalBERT versus the baseline models. The results demonstrate a clear trade-off between predictive accuracy and

processing speed. While InLegalBERT expects more computational resources, its execution time remains well under essential thresholds, making it commercially ready for deployment in legal SaaS environments that need high-throughput automatic judicial analysis.

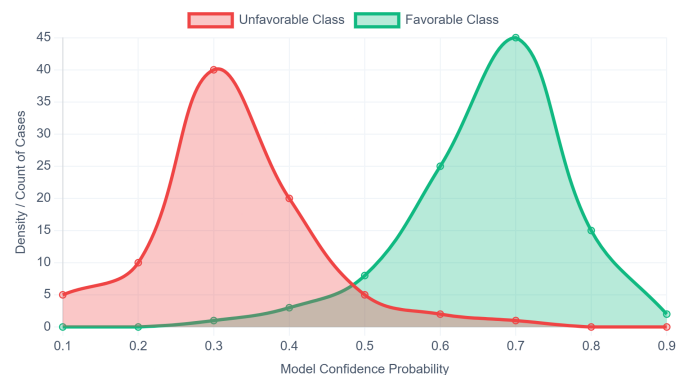


Fig. 12. Prediction Confidence Distribution showing probability distribution of outcomes.

The PCD graph visualizes the model’s discriminative confidence across the test set. By plotting the density of favorable versus unfavorable classifications, the visualization reveals how effectively the InLegalBERT model separates opposing judicial outcomes. The high peaks in the probability bands indicate that the classifier is highly certain in its predictions, effectively reducing the risk of neutral or uncertain decision-making in the automated analysis of Indian legal documents.

B. Interpretability with SHAP

The integration of SHAP successfully resolved the black-box dilemma. The UI renders interactive visual explanations (see Fig. 13) where words contributing positively toward a specific outcome are highlighted in red, and opposing words are highlighted in blue.

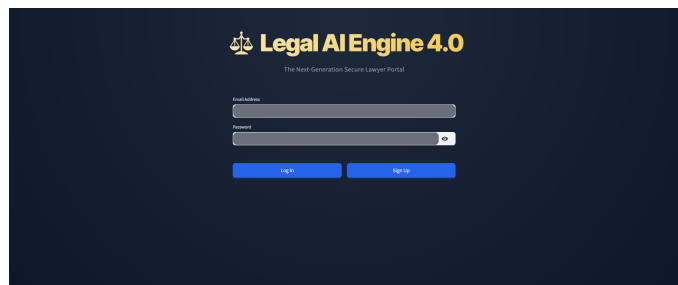


Fig. 13. The platform login interface for secure access.

The protected lawyer threshold interface serves as the principal entrance to the predictive context. The situation features an encrypted authentication system that protects the confidentiality of justice documents and user-specific case histories. By mixing a professional-grade login portal, the system confirms that sensitive legal analysis and AI-generated understandings are restricted to prove practitioners, connecting

the gap between advanced machine learning research and real-world profitable security requirements.

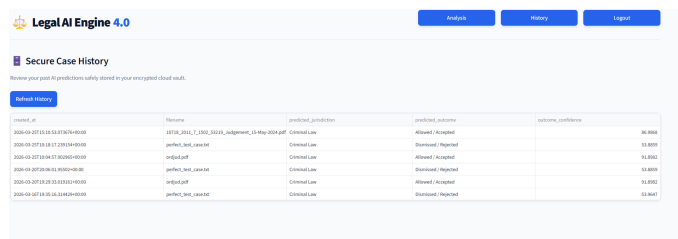


Fig. 14. The Secure Case History dashboard showing historically predicted judgments.

The case history interface functions as a professional audit log, enabling practitioners to retrieve and review previously processed judicial predictions. It integrates seamlessly with the cloud backend to provide real-time updates and persistence for case-specific analytics. This feature is essential for commercial judicial applications, as it allows for the comparative analysis of legal trends and ensures that users maintain a permanent, encrypted record of AI-driven insights for their case preparation strategies.

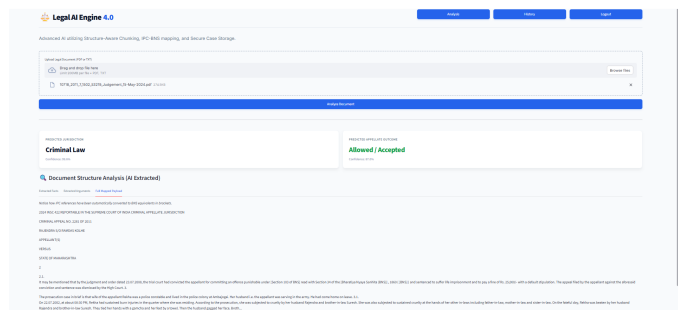


Fig. 15. The detailed case analysis dashboard with real-time prediction results.

The complete analysis interface acts as the primary workspace for legal outcome prediction. It integrates the overview of predictions with in-depth structural analysis, such as court/jurisdiction categorization and legal argument identification. By providing a multi-level perspective of the document’s key legal factors, the platform enables legal professionals to perform a quick audit of the model’s decision-making process. This full transparency is critical for industry adoption, as it ensure that the automated predictions are grounded in the case-specific facts and legal provisions present within the uploaded court judgments.

Our NER module guarantees that the produced visualizations are free from irrelevant stop-words, allowing legal practitioners to quickly validate the model’s reasoning using only extracted legal provisions and case facts.

C. Challenges and Limitations

The primary difficulties encountered was confirming the dynamic JSON mapping properly translated overlapping legal codes without damaging the surrounding text. Additionally,

optimizing large Transformer models on wide text sequences verified computationally expensive. A present limitation of the system is its restriction to binary classification; future iterations seek to expand this to granular multi-class outcomes.

VIII. CONCLUSION AND FUTURE WORK

In this project, we successfully engineered a production-grade, full-stack machine learning system to predict the consequences of Indian legal cases. Through addressing difficult challenges such as the legal evolution since the IPC to the BNS by dynamic middleware, and overwhelming standard transformer token limits via structure-aware chunking, the architecture proves evidence highly strong for modern jurisprudence. Additionally, take part a secure Supabase backend with an collaborating, SHAP qualified Streamlit frontend transforms the theoretical AI model obsessed by a practical, transparent decision-support tool. Our system establishes that Artificial Intelligence has huge, instant potential in transforming legal research and case strategy.

There are quite a lot of exciting opportunities to enhance this raised area in the future. Mixing larger LLMs may possibly qualify generative case summarization along with prediction. Furthermore, drifting the frontend from Streamlit to a fully modified React.js or Next.js architecture would consent for even greater UI/UX tractability and continuous mobile support for enthusiastic lawyers on the go. In conclusion, increasing the dynamic JSON vocabularies to consist of regional state laws would considerably extend the system's general efficacy.

ACKNOWLEDGMENT

The authors would like to thank Mrs. Priyanka Patil for her essential guidance and support throughout this project. We also acknowledge the computational resources and academic surroundings given by the Nutan College of Engineering and Research, which strongly supported the training of our deep learning models.

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